ELEC60008
ELEC70089

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

EEE/EIE PART III/IV: MEng, BEng and ACGI

CONTROL ENGINEERING

Monday, 8 December 2025, 10:00 AM

Time allowed: 3:00 hours

There are THREE questions on this paper.

Answer ALL questions.

Question ONE, TWO, and THREE carry 40, 30 and 30 marks respectively.

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) :

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CONTROL ENGINEERING

Information for candidates:

- The solution of $\dot{x}(t) = Ax(t) + Bu(t)$ is

$$x(t) = e^{A(t-t_0)}x(t_0) + \int_{t_0}^{t} e^{A(t-\tau)}Bu(\tau)d\tau.$$

- Given a matrix A, the matrix exponential is defined as

$$e^{At} = I + At + \frac{(At)^2}{2!} + \frac{(At)^3}{3!} + \dots$$

- The solution of $x(k+1) = Ax(k) + Bu(k)$ is

$$x(k) = A^{(k-k_0)}x(k_0) + \sum_{j=k_0}^{k-1} A^{k-j-1}Bu(j).$$

- The reachability matrix is $\mathscr{R} = [B\ AB\ \dots\ A^{n-1}B]$. A system is reachable if and only if $\text{rank}[\mathscr{R}] = n$. A discrete-time system is controllable if and only if $\text{rank}[A^n\ \mathscr{R}] = \text{rank}[\mathscr{R}]$.
- A system is reachable if and only if $\text{rank}\begin{bmatrix} A-\lambda I & B \end{bmatrix} = n$ for all $\lambda \in \sigma(A)$.
- The observability matrix is $\mathscr{O} = \begin{bmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{bmatrix}$. A system is observable if and only if $\text{rank}[\mathscr{O}] = n$.
  A discrete-time system is reconstructable if and only if $\text{rank}\begin{bmatrix} A^n \\ \mathscr{O} \end{bmatrix} = \text{rank}[\mathscr{O}]$.
- A system is observable if and only if $\text{rank}\begin{bmatrix} A-\lambda I \\ C \end{bmatrix} = n$ for all $\lambda \in \sigma(A)$.
- Ackermann's formula: assume $m = 1$ (single-input system) and let $p_{des}(\lambda)$ be a desired closed-loop polynomial. Then
$$K = -[0\ 0\ \dots\ 1]\mathscr{R}^{-1}p_{des}(A)$$
where $\mathscr{R}$ is the reachability matrix, is such that characteristic polynomial of $A + BK$ is $p_{des}(\lambda)$.

1. Consider the following short questions accounting, overall, for 40% of the exam mark (that is 40 marks out of 100).

a) Consider the discrete-time linear system $x(k+1) = Ax(k)$ with

$$A = \begin{bmatrix} 0.5 & a \\ 0 & -0.8 \end{bmatrix}.$$

Determine the range of values for the parameter $a \in \mathbb{R}$ for which the zero equilibrium point is (i) asymptotically stable, (ii) stable, and (iii) unstable.

[5 marks]

b) Consider the discrete-time nonlinear system described by the equations

$$x_1(k+1) = x_1(k) + \sin(x_2(k)), \qquad x_2(k+1) = x_2(k)\cos(x_1(k)).$$

Determine all equilibrium points of the system. [5 marks]

c) For the system in part (b), write the linearisation around the equilibrium point $\bar{x} = (0,0)$. Assess the stability properties of the obtained linearized system. Then discuss the stability of the zero equilibrium point of the nonlinear system using the principle of stability in the first approximation. [5 marks]

d) Consider the discrete-time linear system $x(k+1) = Ax(k) + Bu(k)$ with

$$A = \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 \\ b \end{bmatrix}.$$

Determine the values of $b \in \mathbb{R}$, if any, for which the system is reachable.

[5 marks]

e) Consider the continuous-time linear system $\dot{x} = Ax(t), y(t) = Cx(t)$ with

$$A = \begin{bmatrix} -1 & 0 \\ 1 & -2 \end{bmatrix}, \qquad C = \begin{bmatrix} c & 1 \end{bmatrix}.$$

Determine the values of $c \in \mathbb{R}$, if any, for which the system is observable.

[5 marks]

f) A continuous-time system described by the equations

$$\dot{x} = Ax + Bu \qquad y = Cx$$

is passive if there exists a matrix $P = P' > 0$ such that

$$A'P + PA \leq 0 \qquad PB = C'.$$

Consider the continuous-time system

$$\dot{x} = \begin{bmatrix} 0 & -2 \\ 3 & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u \qquad y = \begin{bmatrix} 0 & 2 \end{bmatrix} x.$$

i) Show that the system is passive by finding the matrix P corresponding to $A'P + PA = 0$. [5 marks]

ii) Let $u = -Ky$. Write the equations of the closed-loop system and determine the values of K for which the system is asymptotically stable. [5 marks]

iii) The zeros of a system with $x \in \mathbb{R}^n$, $u \in \mathbb{R}$ and $y \in \mathbb{R}$, are the complex numbers $\bar{s}$ such that

$$\text{rank} \begin{bmatrix} \bar{s}I - A & B \\ C & 0 \end{bmatrix} < n+1.$$

A passive system does not have zeros $\bar{s}$ with positive real part. Compute the zeros of the system to verify that these do not have positive real part.

[5 marks]

2. Consider a linear system described by the equations

$$\begin{aligned} \sigma x_1 &= x_2, \\ \sigma x_2 &= u, \end{aligned}$$

with state $x = [x_1, x_2]'$ and input u. Note that one does not know if the system is continuous-time or discrete-time.

a) If the system were continuous-time, assess the stability properties of the zero equilibrium. If the system were discrete-time, assess the stability properties of the zero equilibrium point. [5 marks]

b) Suppose one wishes to design a state feedback $u = Kx$ stabilizing the closed-loop system without knowing if the system is continuous-time or discrete-time. Determine in the complex plane the set to which the eigenvalues of the closed-loop system should belong to ensure to have asymptotic stability of the closed-loop system. [5 marks]

c) Show that the points $\lambda_1 = -\frac{1}{2}$ and $\lambda_2 = -\frac{3}{8}$ belong to the stability region determined in part a) and determine the state feedback gain K such that the closed-loop system, with matrix $A_{cl} = A + BK$, has eigenvalues at $\lambda_1 = -\frac{1}{2}$ and $\lambda_2 = -\frac{3}{8}$. [5 marks]

d) Suppose that the system is a continuous-time system for 1 second, then a discrete-time system for one step, and that this behaviour repeats itself indefinitely. Assume that the discrete-time behaviour is instantaneous, that is the discrete-time evolution occurs at $t = 1$, $t = 2$ and so on. With $x(1)$, $x(2)$, etc. we indicate the state before the jump and with $\bar{x}(1)$, $\bar{x}(2)$, etc. we indicate the state after the jump.

i) Show that

$$\bar{x}(1) = A_{cl} e^{A_{cl}} x(0), \quad \bar{x}(2) = (A_{cl} e^{A_{cl}})^2 x(0), \quad \cdots \quad \bar{x}(k) = (A_{cl} e^{A_{cl}})^k x(0),$$

where A_{cl} is the closed-loop matrix from part c). Hence argue that the state of the system at time $t = k$, with k a non-negative integer, is such that

$$\bar{x}(k+1) = A_{cl} e^{A_{cl}} \bar{x}(k). \qquad (\star)$$

[5 marks]

ii) Determine the eigenvalues of the discrete-time system $(\star)$ determined in part d.i) and show that the discrete-time system is asymptotically stable.
(Hint: you do not need to compute the matrix $e^{A_{cl}}$. Exploit the definition of eigenvalue, that is the fact that if λ is an eigenvalue of A then $Av = \lambda v$ for some non-zero vector v, together with the fact that if λ is an eigenvalue of A, then e^{λ} is an eigenvalue of e^A with the same eigenvector.) [6 marks]

iii) Sketch a plausible time evolution of the variable x_1 for $t \geq 0$.
(Hint: recall that the system behaves like a continuous-time system for all times t which do not have an integer value, and as a discrete-time system when t is an integer. You do not need to show the actual evolution, but just a plausible one. This means that you do not need to compute e^A_{cl} or A_{cl}, nor you need to know any $x(0)$. Sketch just a plausible evolution.)

[4 marks]

3. Consider a linear, discrete-time, system described by the equations

$$\begin{aligned} x_1^+ &= x_2, \\ x_2^+ &= \varepsilon x_1 + 2x_2, \end{aligned} \qquad y = -\varepsilon x_1 + x_2$$

with state $x = [x_1, x_2]'$ and output y. The parameter ε is constant.

a) Write the matrices A and C of a state space description of the system and determine the eigenvalues of A.

[4 marks]

b) Study the stability properties of the system as a function of ε.

[5 marks]

c) Determine conditions on the coefficient ε such that the system is observable. Identify the unobservable mode, if it exists. [6 marks]

d) Using the results in part b) determine conditions on ε such that the system is reconstructable. [5 marks]

e) Compute the observer gain L such that all eigenvalues of the observer error dynamics are at 0. Note that L is a function of ε. Show that $\lim_{\varepsilon \to 0} L$ is well-defined, whereas $\lim_{\varepsilon \to 3} L$ is not. Explain why. [6 marks]

f) Let $L_0 = \lim_{\varepsilon \to 0} L$. Suppose that one uses this observer gain to observe the states of the system for all values of ε (not just for $\varepsilon = 0$). Write $A + L_0 C$ and determine for which values of ε the observer error dynamics is asymptotically stable.

[4 marks]